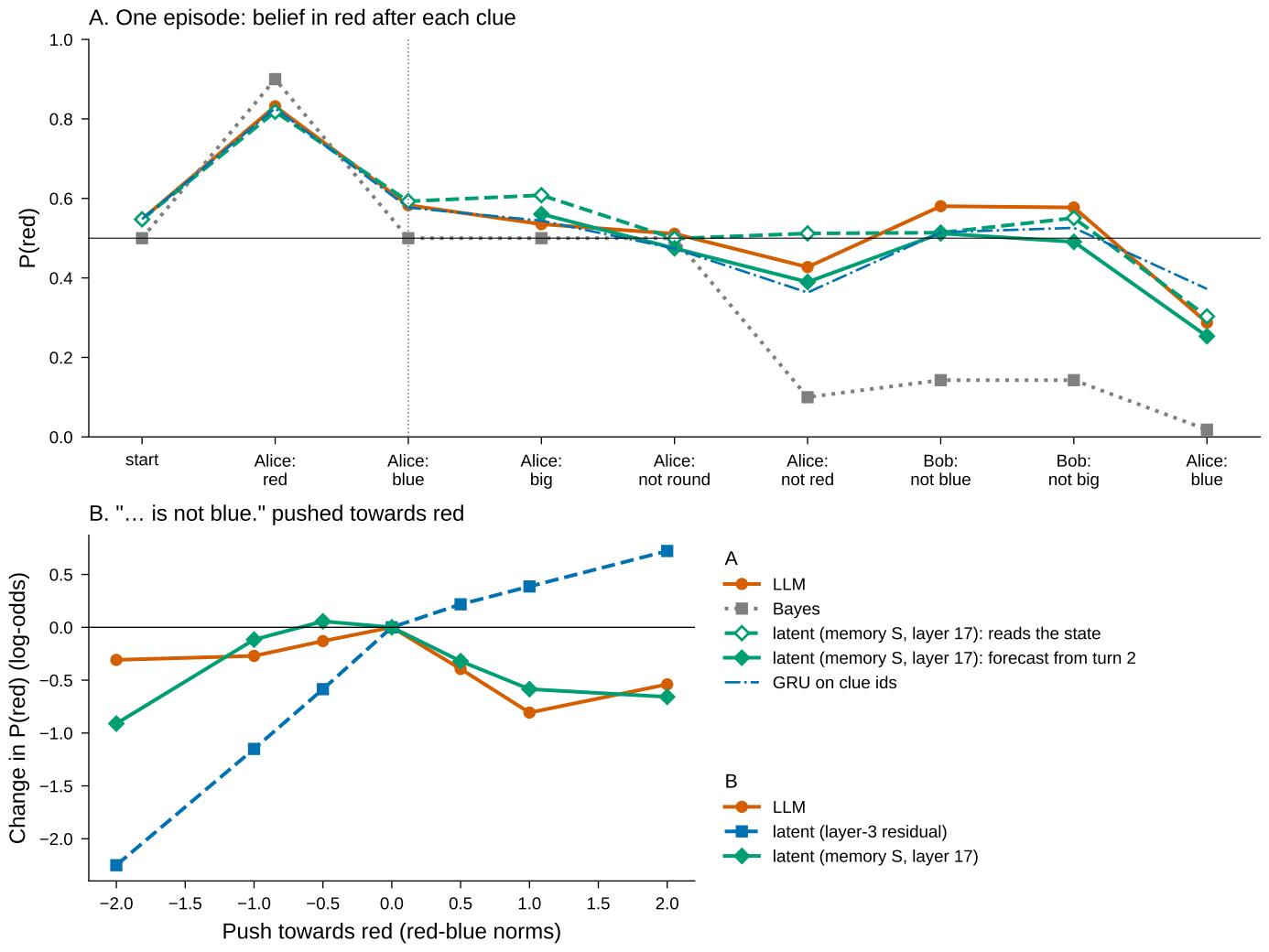
A. One episode: belief in red after each clue
P(red)
1.0
0.8
0.6
0.4
0.2
0.0
start
Alice: red
Alice: blue
Alice: big
Alice: not round
Alice: not red
Bob: not blue
Bob: not big
Alice: blue
B. "… is not blue." pushed towards red
Change in P(red) (log-odds)
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
Push towards red (red-blue norms)
A
LLM
Bayes
latent (memory S, layer 17): reads the state
latent (memory S, layer 17): forecast from turn 2
GRU on clue ids
B
LLM
latent (layer-3 residual)
latent (memory S, layer 17)